

Percolation Weight and Thermodynamic Stability Across a 186-Compound Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

Gilles Frapper
Professor, Theoretical Chemistry
IC2MP UMR 7285 CNRS, Université de Poitiers
gilles.frappier@univ-poitiers.fr

Report generated as part of the `viability` project

August 14, 2026

Contents

1	Objective and Problem Statement	2
2	Methodology	2
2.1	Dataset Selection	2
2.2	Bonding-Type Classification	2
2.3	VASP + LOBSTER Computation	2
2.4	Statistics	3
3	Workflow Diagram	3
4	Environment	3
5	Dataset	3
6	Results	4
6.1	Correlations (Spearman)	4
6.2	Reference Logistic Regression	4
6.3	Figures	5
7	Failures Found and Fixed	5
8	Compute Cost	6
9	Limits and Next Steps	6
10	Conclusion	6

1 Objective and Problem Statement

The first report (6 toy compounds) established that the percolation descriptor computed by `percolation_path.py` diverges sharply from classic ICOHP aggregates (sum, mean, min, max), but without a sample large enough to statistically test whether that difference is *useful*: does the percolation descriptor carry predictive information about thermodynamic stability (`energy_above_hull`) that the classic aggregates don't already carry?

This second campaign builds a 186-compound dataset (180 new + the 6 pilot compounds) split into three families:

- **exp_stable**: experimental (ICSD-referenced) compounds on the Materials Project thermodynamic convex hull;
- **exp_metastable**: experimental metastable compounds ($0 < E_{\text{hull}} \leq 100$ meV/atom);
- **theo_metastable**: theoretical (never experimentally observed) metastable compounds ($0 < \bar{E}_{\text{hull}} \leq 200$ meV/atom).

The `exp_metastable` / `theo_metastable` split serves as a proxy for kinetic persistence: an experimentally observed metastable compound has, by definition, survived long enough to be synthesized and characterized, unlike a theoretical compound of comparable thermodynamic metastability.

2 Methodology

2.1 Dataset Selection

Materials Project query (`mp_dataset/select_campaign.py`): unary and binary chemical systems, non-magnetic ($|\text{total magnetization}| \leq 0.01 \mu_B/\text{formula unit}$ — MP's categorical `magnetic_ordering` field is "Unknown" for ~98% of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements, ≤ 20 atoms/cell, 60 compounds per family with a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by `fetch_candidates.py`'s `classify()` function (**reused unchanged**, per the campaign's instructions), from `is_metal` and elemental composition. This is a secondary analysis axis here, not the primary sampling axis (unlike the campaign's originally proposed plan, replaced at the user's request with the stable/metastable scheme above).

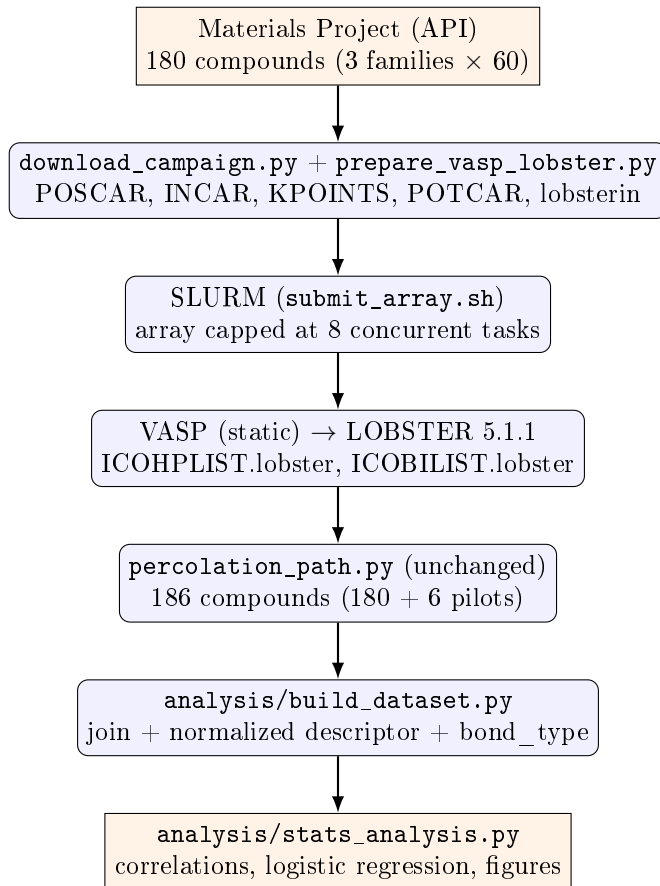
2.3 VASP + LOBSTER Computation

Same template as the pilot campaign (static run, `ISYM=-1`, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`), with two improvements: `NBANDS` is no longer a fixed value but derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 Statistics

Spearman correlation (no reason to expect a linear relationship) between percolation weight (raw and normalized by $|\text{ICOHP}_{\min}|$) and `energy_above_hull`, overall and per bond type; same for the classic ICOHP aggregates, for direct comparison; a reference logistic regression (stable vs. metastable, 2–3 features, 5-fold cross-validation) for a reference AUC. **No SISSO, no symbolic regression**: premature with a single candidate descriptor at this sample size (see §9).

3 Workflow Diagram



4 Environment

Same as the pilot campaign (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib` (see `requirements.txt`).

5 Dataset

Family	n	E_{hull} range (eV/atom)
exp_stable	63	0.0000 (by construction)
exp_metastable	63	0.0005 – 0.0907
theo_metastable	60	0.0013 – 0.1991

Bond type (<code>classify()</code>)	<i>n</i>
metallic	88
covalent	23
ionic	9
unclassified	66

35% of compounds (66/186) fall outside `classify()`'s three categories — mostly transition-metal pnictides/chalcogenides and hydrogen-containing compounds, a real limitation of the reused heuristic, documented but not fixed here.

6 Results

6.1 Correlations (Spearman)

Group	Metric	<i>n</i>	ρ	<i>p</i>
all	percolation weight (raw)	186	0.064	0.383
all	percolation weight (normalized)	186	0.071	0.339
metallic	percolation weight (raw)	88	0.191	0.075
metallic	percolation weight (normalized)	88	0.222	0.038
ionic*	percolation weight (raw)	9	−0.402	0.284
covalent	percolation weight (raw)	23	0.165	0.453
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

* *n* < 15: do not over-interpret.

In the one subgroup where any result reaches nominal significance (metallic, *n* = 88), the normalized percolation weight (ρ = 0.222, *p* = 0.038) and the ICOHP sum (ρ = 0.223, *p* = 0.037) are statistically indistinguishable in strength. **The percolation descriptor does not demonstrate additional predictive power over the classic aggregates on this dataset** — a more precise answer than the previous (6-compound) report's "qualitatively different" framing, and a more directly actionable one.

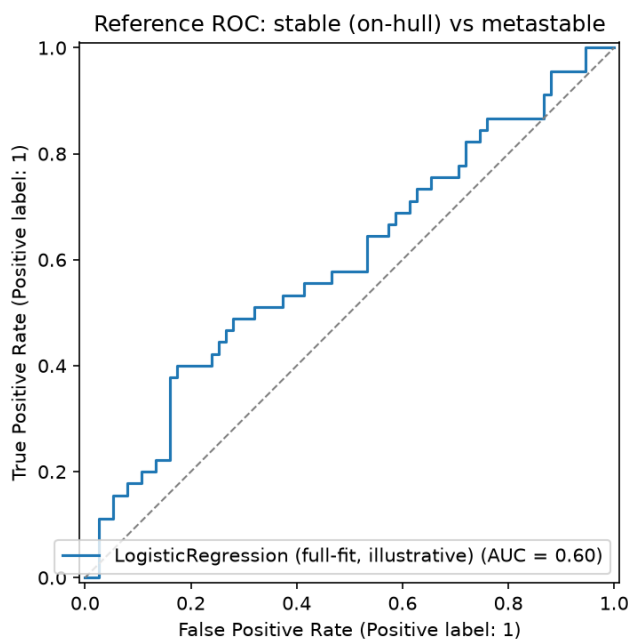
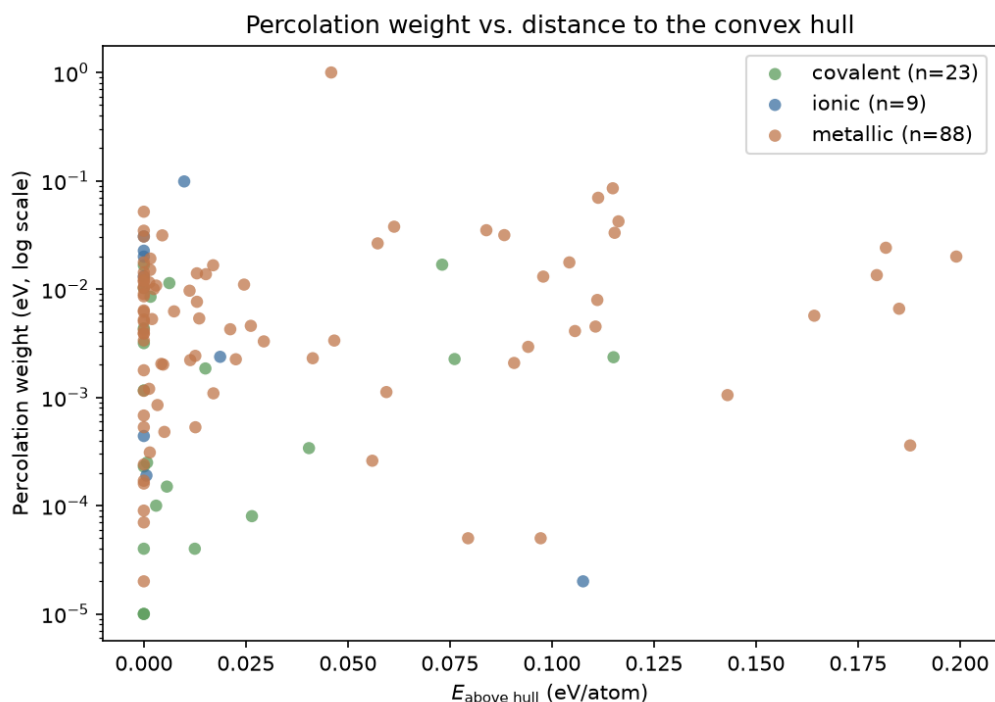
6.2 Reference Logistic Regression

Stable (on-hull) vs. metastable, features: normalized percolation weight, mean ICOHP, bond-type dummies (*n* = 120 after dropping unclassified compounds).

AUC (5-fold cross-validation)	0.496 ± 0.141
per-fold AUC	0.556, 0.430, 0.252, 0.637, 0.607

AUC $\approx$ 0.5: chance level. The wide spread across folds (0.25–0.64) is itself informative: it indicates the absence of a stable, learnable signal at this sample size, not merely a "weak but real" one.

6.3 Figures



7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source:

- **Al12W, BW, WO2, TeW, TiW** (all tungsten-containing): the local mirror's `W_sv` POTCAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).

- **A112W** failed again after that fix, this time with a VASP SIGSEGV — exactly the known issue flagged in the mission brief (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`'s per-compound SLURM template. Fixed; recovered on retry.

Final success rate: **186/186**.

8 Compute Cost

	VASP	LOBSTER
Mean (s)	124	397
Median (s)	72	145
Max (s)	1440	6675
Sequential total	6.4 h	19.1 h

Actual elapsed wall-clock time (8-way concurrency): ~ 3.5 h, within the 6 h overnight budget.

9 Limits and Next Steps

- **Sample size:** 186 overall, but 88/23/9 once split by bond type — the "ionic" correlations ($n = 9$) are not interpretable.
- **Primitive vs. conventional cell** (carried over from the pilot report): not addressed here. Most likely explanation for the weak observed correlations — the strongest bond can sit entirely inside the primitive cell and contribute to no non-contractile cycle at all, biasing the measured "weakest link."
- **No true dynamical "unstable" class:** `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.
- **classify() coverage:** 35% of compounds unclassified.
- **Why not SISSO yet:** a single candidate descriptor (the percolation weight, in two algebraically related variants), compared against aggregates that are themselves functions of the same underlying ICOHP values — no meaningful combinatorial search space yet. Justifying SISSO would require: (a) several conceptually novel candidate descriptors, (b) per-stratum $n \geq 50$ –100, (c) a clearer signal than $\text{AUC} \approx 0.5$.

10 Conclusion

The 186-compound dataset was built, computed (VASP+LOBSTER), and analyzed end to end, with two compute failures diagnosed and fixed at the source. The headline result is a *negative* but quantified and honest one: at this sample size, the percolation descriptor shows no significant correlation with thermodynamic stability, and demonstrates no advantage over classic ICOHP aggregates. The priority actions (§9) — especially conventional-cell reprocessing and a targeted extension of the metallic subsample — are likely to move this result before a SISSO-class tool is warranted.

Source code: <https://github.com/John-Akapulco/viability>

Data: `analysis/percolation_vs_hull.csv`

Detailed report: `analysis/REPORT.md`